

Article

Integrating Generative AI in Higher Education: Teachers' Perceptions Through the TPACK Lens

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Abstract

Generative AI (GenAI) in education has become a divisive topic. While some teachers view GenAI as a transformative tool, others caution against harms and impacts on teaching and learning. Since the usefulness of a tool depends on the user, how teachers view and use GenAI and their perceptions of its role in teaching and learning may influence its benefits and risks. The study adopts a qualitative approach, interviewing 23 higher education teachers with teaching experience across disciplines. Using the Technological, Pedagogical, and Content Knowledge (TPACK) and sensemaking frameworks to guide and analyse the interviews, a set of opportunities, challenges, and threats associated with integrating GenAI in higher education was identified. Teachers highlighted opportunities, including using GenAI as a study buddy. Concerns were raised about students' overreliance on GenAI, GenAI undermining the teaching and learning process, and issues like undetected plagiarism. The findings suggest a need for professional development to help teachers understand GenAI and how it can be effectively used in teaching. Some teachers warned against the paradox of using a tool to save time, only to find that it might increase workload and frustration. These insights contribute to developing guidelines and informing policymaking to ensure integration of GenAI in education.

Keywords: generative AI; TPACK; teachers' perceptions; higher education; professional development

1. Introduction

The emergence and rapid adoption of generative artificial intelligence (GenAI) have elicited polarized responses in higher education, characterized by both enthusiasm and concern among educational stakeholders, including administrators, teachers, and students (den Hollander et al., 2025; Kim et al., 2025; Obenza et al., 2024). While advocates view GenAI as a transformative tool capable of enhancing teaching and learning, critics raise concerns about its potential negative effects, such as undermining students' critical thinking skills and posing a threat to academic integrity (Yan et al., 2024). As institutions attempt to reconcile these tensions, teachers remain key actors in implementing GenAI in practice. Therefore, understanding how teachers perceive opportunities and challenges associated with integrating GenAI is crucial.

Teaching is a dynamic profession that continually evolves in response to technological and societal changes. Throughout history, as new tools have created new opportunities for teaching and learning (e.g., from textbooks to instructional videos), teachers have adapted their teaching practices to incorporate these technologies in ways that align with



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their instructional goals (Reiser, 2001). Although AI has long been used in education in various forms, such as intelligent tutoring systems and learning analytics dashboards (Ouyang & Jiao, 2021), the rapid adoption of GenAI has placed AI in the spotlight of public and academic attention. As the term suggests, GenAI, largely powered by large language models, generates content as text, images, audio, or video that resembles human-created output. A prominent application of GenAI is chatbots, such as OpenAI's ChatGPT, Microsoft's Copilot, Anthropic's Claude, and Google's Gemini, designed to interact through natural language in human-like ways, such as by asking follow-up questions, admitting mistakes, and rejecting inappropriate requests (Giannakos et al., 2025). While GenAI was not initially developed for educational purposes, its capabilities are increasingly being integrated into teaching and learning contexts.

GenAI holds considerable promise for education by scaling personalized support, diversifying learning resources, enabling timely feedback, and innovating assessment practices, while simultaneously raising ethical concerns about transparency, privacy, equity, and assessment practices (Yan et al., 2024). Whether these opportunities are realized and challenges addressed depends partly on teachers' professional knowledge on integrating GenAI into their teaching practice. While several studies have examined teachers' perceptions of GenAI (e.g., Bower et al., 2024), there remains a gap in our understanding of how these perceptions are influenced by teachers' professional knowledge.

The Technological Pedagogical Content Knowledge (TPACK; Mishra & Koehler, 2006) framework describes multiple forms of teacher knowledge and their integration, where changes in one form affect the others. For example, teachers' content and pedagogical knowledge influence how they use GenAI (Feldman-Maggor et al., 2025). These various forms of knowledge operate within teachers' understanding of broader systemic factors that influence teaching and learning. Extending TPACK, recent studies have proposed additional relevant forms of knowledge in the context of GenAI, including ethical knowledge and human–AI collaborative knowledge (Celik, 2023; Chiu, 2025). However, from a sensemaking perspective, the implementation of technological tools in education is not a straightforward process. New technologies, such as GenAI, often introduce uncertainty and ambiguity, leading educators to consider and reflect on their potential risks, advantages, and overall impact on existing teaching practices. Rather than simply applying existing knowledge, teachers actively interpret new technologies through their beliefs, experiences, and professional knowledge related to technology, subject matter, and pedagogy (Kaplan-Rakowski et al., 2023).

Sensemaking in this study acts as a process-oriented lens to help us better understand teachers' interpretations of the opportunities and challenges associated with GenAI and how these interpretations form the ways different domains of knowledge, such as technological, pedagogical, and ethical knowledge, are stimulated and integrated. This process of interpretation often takes place within institutional contexts, where educators exchange their perspectives with colleagues and react to evolving educational policies and expectations around GenAI integration (Georgiou et al., 2025). Thus, through sensemaking that complements the TPACK framework, we first highlight how teachers construct meaning around GenAI and then emphasize how this process of meaning making influences the way teachers apply and integrate their professional knowledge in practice.

Insights into teachers' perceptions of GenAI in higher education can provide a deeper understanding of how beliefs, professional knowledge, and institutional factors may influence the use of GenAI for teaching, learning, and assessment. Therefore, using the TPACK framework in combination with the sensemaking framework, we explore higher education teachers' perceptions of GenAI and how these perceptions shape their pedagogical decision-making.

This study contributes to the literature on GenAI in higher education in several ways. First, we integrate the TPACK framework to explore teachers' domains of professional knowledge while also incorporating a sensemaking perspective to examine how teachers interpret and mobilize this knowledge when engaging with GenAI. Second, through rich qualitative insights, we give voice to higher education teachers to better understand how challenges, concerns, and opportunities related to GenAI are linked to teachers' professional knowledge. Finally, the findings offer implications for professional development and institutional policy to support teachers in making informed pedagogical decisions when using GenAI.

1.1. Integrating Generative AI into Teaching Practices

Research is emerging to understand how teachers are engaging with GenAI tools. For example, [Bent et al. \(2025\)](#) analyzed higher education instructors' conversations with Claude and found that curriculum development was the most prevalent use, followed by academic research and the assessment of student performance. Additional applications included creating mock scenarios for educational simulations, developing training content, drafting recommendation letters, and creating meeting agendas or administrative documents. Overall, instructors primarily used GenAI to automate time-consuming tasks, act as a collaborative thought partner for idea generation, and enable a more personalized learning experience by creating individualized interactive activities.

While [Bent et al.'s \(2025\)](#) study offered insights into how teachers used a specific GenAI tool, [Bower et al. \(2024\)](#) employed a broader and more international approach to examine teachers' perceptions of the impact of GenAI on teaching and assessment across diverse teaching levels, experience levels, and disciplines. Their large-scale mixed-methods study revealed that perceptions of teachers vary by awareness of GenAI, teaching experience, teaching level, discipline, region, and gender. Specifically, teachers with greater awareness of GenAI tools perceived the impact of these tools as more profound or major. The findings emphasize the importance of considering contextual and teacher-level differences when examining the nuanced impact of GenAI on teaching practices. At the same time, the results point to a gap between teachers' perceptions of how teaching and assessment should change, and the themes proposed in recent research on how AI can be used to enhance teaching and assessment ([Chiu et al., 2023](#)), such as providing feedback, monitoring student progress, and AI-assisted assessment. This gap may reflect limited teacher knowledge about GenAI's full potential and a corresponding need for professional development.

Further evidence of this need is provided by [Kaplan-Rakowski et al.'s \(2023\)](#) study, which showed that teachers' greater awareness of GenAI was linked to more frequent use. Furthermore, teachers' frequency of GenAI use was associated with their stages of GenAI integration. Approximately half of the teachers surveyed reported that they could think of specific tasks for which GenAI might be useful (understanding stage) and felt comfortable in using GenAI for specific tasks (familiarity stage). These findings suggest that increased awareness may support more frequent GenAI use and, consequently, progression towards higher stages of GenAI integration, including adaptation and creative application. While teachers recognize the ease of use of GenAI and its potential to save time, they also expressed the potential impact on students' critical thinking and academic integrity. Therefore, despite generally positive perceptions, teachers remain cautious about how to meaningfully integrate GenAI into their teaching practice.

Concerns about institutional support also highlight the need for targeted professional development. [Barrett and Pack \(2023\)](#) found that a vast majority (90%) of the higher education instructors reported that their institution lacked a clear GenAI policy and that they had not received training on the use of GenAI. Similarly, most instructors reported

not having taught their students about the appropriate use of AI. The results suggest that, in addition to the need for institutions to establish clearer policies on the ethical and responsible use of GenAI, educators require training to develop skills and competencies to use GenAI tools more proficiently and to guide students in its appropriate use.

Although these studies provide valuable insights into teachers' perceptions and current practices regarding GenAI, there remains a limited understanding of the professional knowledge teachers possess and their need to integrate GenAI effectively into their teaching practice. Little is known about how this knowledge is shaped by the interaction of technological, pedagogical, and content knowledge, as conceptualized in the TPACK framework.

1.2. TPACK in the Context of Generative AI

TPACK (Mishra & Koehler, 2006) is a highly influential model for understanding the knowledge teachers require to effectively integrate technology into their teaching practice. Building on the premise that teaching is a highly complex activity, TPACK proposes that meaningful use of technology to enhance teaching depends on the intentional integration of three main sources of knowledge, namely technology, pedagogy, and content. In the traditional version of TPACK, four main technology-related areas are described: (1) Technological Knowledge (TK) and its intersections with pedagogy and content, forming (2) Technological Content Knowledge (TCK) and (3) Technological Pedagogical Knowledge (TPK), which converge to form (4) TPACK. Researchers have argued that the traditional TPACK framework is insufficient in GenAI contexts, as GenAI does not merely represent another digital tool, but a transformative socio-technical phenomenon that requires teachers to challenge existing assumptions about teaching and learning (Celik, 2023; Chiu et al., 2026). However, the TPACK framework can be considered technology-agnostic (Mishra et al., 2023), offering a flexible lens for examining teachers' knowledge needs across its core components.

Recent research shows that TPACK plays an important role in teachers' willingness to integrate GenAI into their teaching practices (Sun et al., 2024; Xia et al., 2025; Yang et al., 2025). However, this relationship is not just direct, but also interacts with cognitive, affective, and motivational factors. For example, higher TPACK leads to greater perceived ease of use, perceived usefulness, and self-efficacy, and in turn, greater willingness to integrate AI in STEM education (Sun et al., 2024). Similarly, teachers with higher TPACK and greater perceived satisfaction of their needs for competence and relatedness are better at overcoming negative emotions (e.g., anxiety) of GenAI integration (Yang et al., 2025). These findings underscore the importance of employing TPACK as a foundational framework to examine teachers' perceived opportunities and challenges with GenAI. Understanding these perceptions can help identify specific professional development needs and inform the design of targeted support programs.

To effectively integrate GenAI into their teaching practices, teachers first need knowledge of the technology itself (i.e., TK), that is, to understand the nature of GenAI and what distinguishes it from other technologies. Mishra et al. (2023) stressed that while GenAI shares characteristics of being protean, unstable, and opaque with other digital technologies, and these characteristics are magnified by its ability to interact across various formats, ranging from text to images to video, via a highly accessible and intuitive natural language interface. Besides these characteristics, GenAI can be distinguished from other technologies by its generative and social nature, producing novel, human-like responses in real time. These characteristics underlie how GenAI may reshape teaching and learning, positioning it as more than a tool but as a collaborative, social "psychological others" that can support humans in complex tasks, while also being prone to errors. Understanding these unique

characteristics is essential for teachers to effectively leverage GenAI's capabilities while mitigating potential risks of bias, hallucinations, and over-reliance.

In addition to knowing the technology itself, teachers need to be able to integrate this knowledge with their pedagogical (i.e., TPK) and content (i.e., TCK) knowledge. TPK refers to teachers' understanding of how GenAI can be used to support pedagogical processes, such as using GenAI to address challenges students might face or new approaches to assessment. An important aspect of TPK includes knowing the appropriate and inappropriate uses of GenAI and supporting learning in acquiring this knowledge as well (Mishra et al., 2023). In contrast, TCK focuses on how GenAI may reshape specific content areas by redefining what disciplinary knowledge in the context of GenAI entails. For example, when students leverage GenAI to generate and debug code, curricula may need to place greater emphasis on critical and computational thinking and strategic problem solving (Viberg et al., 2025). Consequently, teachers need to reconsider what forms of domain-specific knowledge remain fundamental and whether learning objectives need to be redefined to prioritize higher-order thinking skills, and how GenAI can be used to enhance students' learning within the discipline.

Besides TK, TPK, and TCK, Mishra et al. (2023) further highlight the need to consider Contextual Knowledge (XK), which encompasses understanding broader systemic factors that shape the integration of technology. Teachers operate within educational systems that impose various constraints over which they have limited control, including institutional policies, national regulations, and the availability of teaching resources and support. Therefore, effective integration of GenAI requires teachers to understand how these contextual factors may shape both possibilities and limitations of what they envision for their teaching practice.

1.3. Contextual Factors: Ethics, Policies in Educational Practice, and Professional Development

Following the rapid entrance of GenAI in higher education, several responses, including policy documents and institutional statements, have emerged, but most responses remain fragmented or unsystematic (Chan, 2023). Uncertainty about how to incorporate GenAI responsibly while safeguarding academic integrity remains a concern for universities and educators. Thus, teachers are often required to make sense of how to use GenAI without clear and consistent institutional guidance.

To address the uncertainty, a few efforts have been made focusing mainly on academic integrity. For example, Oxford and Cambridge, together with six more universities out of the twenty-four universities in the UK's prestigious Russell Group, classified the use of GenAI for assignments as academic misconduct. At the same time, several Australian institutions have reintroduced pen and paper assessments in response to concerns about academic integrity (Barrett & Pack, 2023; Chan, 2023). These responses, however, reflect institutions' efforts to prioritize risk management over pedagogical integration, leaving educators to navigate tensions between student learning, innovation, and integrity in their own teaching practice.

At the policy level, universities have largely emphasized ethical principles, developed by organizations including UNESCO and the European Union, following frameworks like Floridi and Cowls (2021) five ethical principles. These principles stress human-centered design (Infocomm Media Development Authority & Personal Data Protection Commission, 2020), legal compliance, and societal robustness (UNESCO, 2021) but fail to address practical, discipline-specific challenges teachers face when integrating GenAI into teaching and assessment (Dexe & Franke, 2020). Assessment strategies, the transformation of the teacher role, equity of access, and infrastructure limitations are only a few of the concerns educators and policymakers identify in education, where this policy gap is particularly evi-

dent (Pelletier et al., 2021; Popenici & Kerr, 2017; Swiecki et al., 2022). From a sensemaking perspective, these concerns influence how teachers make sense of institutional expectations and decide if, how, and how much to use GenAI in their classrooms.

While educators are at the frontline, this policy-practice gap may still lack the necessary institutional support and preparation to respond properly. Professional development efforts that offer hands-on tools and continuous support may help educators to integrate GenAI meaningfully in their teaching (Paek & Kim, 2021). However, existing professional development initiatives often focus narrowly on plagiarism detection or are slow to include proper GenAI training and neglect meaningful instructional uses of GenAI (Weiner et al., 2025).

In a nutshell, in higher education, there is a clear misalignment between institutional policies, ethical frameworks, and the realities of teaching and learning practice. Understanding how teachers navigate the challenges through their sensemaking processes, their professional knowledge, and beliefs may be essential for the development of professional development initiatives and institutional guidelines that support ethical and effective GenAI integration.

1.4. Current Study

This study explores how higher education teachers perceive and use GenAI tools in their teaching practice, and how their perceptions shape their pedagogical decisions. We use the TPACK and sensemaking frameworks as lenses to explore teachers' technological, pedagogical, and content knowledge, their acceptance of GenAI, and how they conceptualize opportunities, challenges, and threats associated with its use. In addition, to better support the development of practice-oriented guidelines that enable meaningful integration of GenAI in higher education, we further explore teachers' views about ethical guidance, professional development needs, and student support. Thus, the study addresses the following research questions:

1. What are teachers' understanding, use, and acceptance of GenAI for their teaching practices?
2. How do teachers conceptualize opportunities, challenges, and threats of GenAI in their teaching practice, and how do these relate to their content and pedagogical knowledge?
3. What type of support and professional development do teachers perceive as necessary for ethical and responsible integration of GenAI in teaching practice?

2. Materials and Methods

2.1. Participants

Participants were 23 university teachers (17 men, 6 women) working at various universities in the Netherlands. The sample represented a diverse range of ages ($M = 46.17$, $SD = 10.50$) across multiple disciplines and institutional contexts, including universities of applied sciences and research universities. In detail, participants came from STEM-related disciplines (e.g., computer science, electrical engineering, civil engineering, business informatics, plant science), social sciences, education and governance (e.g., psychology, teacher education, social work, educational sciences, economics), humanities (e.g., history, religious studies, Islamic studies), and health and applied professional fields (e.g., dietetics, nursing). Participants had varying levels of teaching experience ($M = 12.17$, $SD = 8.74$). Participants were recruited through multiple channels, including personal networks and social media platforms.

2.2. Semi-Structured Interview Guide

A semi-structured interview guide comprising 9 main open-ended questions with sub-questions was developed. Questions were derived from the Intelligence-TPACK questionnaire (Celik, 2023) and relevant studies (Koehler & Mishra, 2009; Mishra & Koehler, 2006). Example items include questions to gain insight into teachers' understanding of what GenAI is (TK): *What is GenAI to you? And which GenAI tools do you know and use?* For TPK, questions were formulated to explore the influence of AI on teaching strategies: *From your perspective, what opportunities or benefits do you think GenAI can offer to support students or improve learning outcomes?* TCK questions examined the interplay between GenAI tools and subject-specific content: *How might GenAI affect your teaching approaches for the type of content you teach?* For TPACK, questions were formulated more generally to assess teachers' overall views: *Can you describe one or two ways in which you are most likely to use Gen AI?* In addition, ethical perspectives were addressed by including questions on ethical implications, for example, *what do you think are potential ethical implications of integrating AI-based tools into education?* The interview guide is included as a Supplementary File.

2.3. Procedure

To ensure that the interview questions were relevant and appropriately communicated, we conducted a pilot study with four university teachers. Participants provided oral feedback on each interview question and reflected on potential improvements. Following the pilot, only minimal adjustments were made for clarity and to avoid repetition. The interview guide was initially formulated in English and then translated into Dutch by two native Dutch speakers. A final consistency check was conducted to ensure accuracy. For the main study, participants were required to complete an informed consent form as a first step prior to the interviews. Interviews were conducted by two student researchers in either Dutch or English, depending on participants' preferences. The interviews lasted between 30 and 60 min. Data collection took place between April and May 2024.

2.4. Data Analysis

All recordings were transcribed verbatim using Amberscript. The transcribed interviews were coded using ATLAS.ti software. Thematic analysis was employed to analyze the data following Braun and Clarke's (2006) framework. The analysis combined both deductive and inductive approaches. First, deductive codes were derived from the conceptual framework of the study and research questions, including TPACK domains, sensemaking, challenges, opportunities, and ethical considerations. Inductive coding, at the same time, allowed new codes to emerge from recurring patterns in the interviews that were not initially captured from the initial framework. The analysis began with data familiarization and the generation of initial codes based on response content and its relevance to the research aim. At this stage, both deductive and inductive codes were grouped into broader themes, which were then reviewed, refined, and defined. To ensure coherence with the TPACK model, the themes were further analyzed in relation to its framework. This process led to the final organization of themes around opportunities and challenges related to TCK and TPK, as well as ethical concerns, beliefs, attitudes, and needs. Subthemes also emerged from the data, such as *knowing what* and *knowing how*, within the broader theme of TK.

The three authors of the study served as coders for the analysis. Trustworthiness was ensured through several measures. To enhance credibility, participants were asked to confirm the accuracy of their transcribed interviews through member checking (Birt et al., 2016). Minor revisions and additional comments were incorporated into the transcripts based on participant feedback. To enhance confirmability (Stenfors et al., 2020), the three coders independently coded the transcripts before comparing their coding. Then, in

three separate coding sessions, coders discussed differences and discrepancies against the data and the study's coding framework. When disagreements occurred, the coders systematically reviewed and discussed all relevant transcript segments and refined code definitions until consensus was reached. This systematic process also resulted in updated and refined versions of the codebook to ensure consistent application of codes across the dataset.

3. Results

Below, we present the findings on how higher education teachers perceive and use GenAI in their teaching practice. To provide a clear organization of the themes, we have structured the results around three interconnected themes: teachers' perceptions and usage of GenAI, perceived opportunities and challenges for teaching and learning, and requirements for responsible and critical integration. The first theme focuses on teachers' understanding of GenAI, their current engagement with GenAI tools, and the extent to which they accept its use. The second theme explores perceived benefits for both teaching practice and student learning, highlighting ways in which GenAI can support pedagogical and content-specific knowledge. The third theme addresses requirements for ethical, inclusive, and effective use, including professional development needs, institutional support, and critical engagement strategies. Quotations are presented verbatim to capture the authenticity of participants' voices, including grammatical inaccuracies.

3.1. Teachers' Understanding, Use, and Acceptance of GenAI

Teachers' understanding of GenAI ranged from basic awareness to more advanced conceptual understanding and applications. While all participants could name at least one GenAI tool, with ChatGPT being the most referenced GenAI tool, others also mentioned the use of other tools, such as Gemini, Copilot, Mistral, and Bing. However, not all participants could clearly describe what GenAI is. Some described it with broad, general terms, calling GenAI a "catch-all" [P03] and "collective" [P23] term for anything a computer can generate. One participant metaphorically described it as a "calculator for words or for pixels" [P22], while another defined GenAI as an "answer machine" [P10] to be contrasted with a search engine. The difference was further elaborated by the participant describing the reduced control one has over the information they receive in comparison to search engines, since GenAI does not provide a set of answers from different websites but instead just one answer. Participants with a more advanced understanding mentioned large language models and attempted to make distinctions between predictive and generative AI, indicating higher levels of technological knowledge.

In terms of usage, several participants described using GenAI tools in exploratory ways to find out what their students will encounter when they start using such technologies. For example, one participant noted, "I'm deliberately only using the free versions, because I actually want to see what students have access to . . . and most students only have access to free versions" [P17]. Similarly, another participant explained, "I thought it was really nice to see what the students can do with it and then to be aware of that, so that you can also form an image of what you want, what you want students to participate in or not to participate, and if they do something with it. How do you want them to report on that?" [P20]. For another participant, such exploration of GenAI tools served as a way to educate themselves on the tool's capabilities and limitations.

Besides exploration, participants also described using GenAI for various educational tasks. One such task concerns creating exam questions, although for one participant, this was still at an exploration stage. One participant compared the use of GenAI to how publishers provide test banks, stating that teachers do not have to do everything

by themselves. Other uses included generating images for presentations and acting as a “sparring partner” during lesson preparation. For example, one participant mentioned, “I have certain topics that I want to discuss, and then I put that in ChatGPT, and then it indicates to me very quickly what it entails, and I can then use that for my preparation for the lessons.” [P18]. Another participant expressed interest in utilizing GenAI tools more to develop new courses or lessons. In-class use was mentioned rarely. One participant incorporated GenAI use into a class assignment with the intention of supporting students’ critical reflection around argumentation. Another described using GenAI together with a student to create a thesis title. More informal applications included making poems in class.

A few participants raised the advantage of efficiency when mentioning the use of GenAI tools for educational task-related purposes. As one put it, “I use it more for speed to quickly map something out and not to really learn something substantive from ChatGPT, because I just notice that it often goes wrong.” [P18]. However, not all experiences were positive. One participant explained their non-use as being due to the effort involved, as it “takes too much energy for me to think about that on the front end, and then I get something out at the back that I thought I didn’t ask for. And then it ultimately takes less energy to just do it yourself.” [P05]. Others noted that they have not yet used GenAI because they did not see the added value in their current teaching context or how students can benefit from it. For example, “So, currently, for what I teach, I don’t see how students could benefit from it or use it. And therefore, also for me, the curiosity is not that high for using it.” [P12]. Another participant made a remark that they are thinking about it and are still searching for potential applications.

3.2. Perceived Opportunities and Challenges of GenAI for Teaching and Learning

3.2.1. Opportunities for Teaching Practice

Across the interviews, teachers highlighted how GenAI can support and enrich teaching preparation and delivery. Several participants described using GenAI as a source of inspiration for lesson planning, generating ideas, and improving classroom activities. One teacher noted, for example, that it can be “a nice source of inspiration. . . and “generate ideas for an active teaching method” [P3]. Others highlighted that GenAI could help make lessons more engaging or fun by offering creative suggestions or illustrative materials.

Several participants also described using GenAI for routine, preparatory, or administrative tasks, such as summarizing meeting notes or identifying action points from a meeting, even reducing “mistakes”. In addition, GenAI was perceived as a brainstorming or sparring partner in designing or calibrating assessments or developing rubrics. Another teacher explained how GenAI could simplify assessment design: “If I just say: well, I want a five-point rubric now and these are five assessment criteria, then something very beautiful will come out of it and I don’t have to do all that myself anymore.” [P3].

In line with this, several participants perceived workload reduction as a key advantage of GenAI for teachers. As one teacher explained, “there can be a certain workload relief. . . You can outsource a little more, and then you can save time. So, I see advantages in that as well. . . it can make the work easier in the long run.” [P21] Importantly, this efficiency was not framed as an end, but rather as creating space for other core aspects of teaching. Another teacher reflected that the time saved could allow teachers to give more attention to students’ well-being and learning processes. In detail, the teacher mentioned that “it doesn’t have to take a lot of time to plan a lesson, leaving more emotional and mental space to attend to students and reflect on what the lesson is doing to them.” [P17] Taken together, we understand that teachers primarily relate opportunities for teaching with technological–pedagogical knowledge, especially in relation to preparation, efficiency, and reflective teaching practice.

3.2.2. Opportunities for Student Learning

Most teachers identified several opportunities for students' learning and did not view GenAI as a tool that necessarily needs to be prohibited, if students are taught how to use it critically and carefully. As one teacher stated, "It's just one tool that you have to teach students to use carefully—with the emphasis on carefully." [P14].

One frequently mentioned opportunity concerned the practical study support that GenAI can offer students, such as helping with summaries or study planning, generating practice questions, brainstorming ideas, and providing help or feedback as a tutor. Interestingly, several participants emphasized that such support could contribute to reducing educational inequalities, especially for students who have no access to academic support in their immediate social network. This was seen as especially valuable for students on the autism spectrum. One teacher reflected:

We have a large group of students who have the autistic spectrum and who therefore do not even dare to ask a fellow classmate or a teacher. And they now have a buddy or an extra classmate they can approach 24 h a day, seven days a week. [P22]

Teachers also highlighted the potential of GenAI to support skill development. Especially domain-general skills like critical thinking or "training your critical eye" [P14] and writing were emphasized several times, along with skills such as asking questions and integrating information. One participant emphasized that "writing should not be outsourced but taught as a way of thinking, more like an active process of organizing, selecting, and creating." [P17].

In addition to these general skills, some opportunities in relation to discipline-specific skills that reflect technological content knowledge were addressed. For example, in engineering, GenAI was seen by one of the participants as a valuable tool for helping students unravel complex "wicked problems" by prompting the right questions to narrow down the problem and consider different solution pathways. One participant in the civil engineering industry explained that such tools can make abstract challenges, such as sanitation, urbanization, and infrastructure shortages, more tangible, supporting students in thinking through issues like limited water or road capacity [P9].

Another participant noted that GenAI is especially useful for tasks involving domain-specific knowledge, where established rules and procedures exist. For example, in nutrition and patient care, GenAI allows students to quickly retrieve relevant guidelines needed to support decision-making in this context:

So, what do you do in case someone has a certain disease? What are the dietary requirements? What about protein? What about oil advice . . . fats? Those kinds of regulations, and those procedures, are easy to look up via AI, so to speak (. . .) Yes, so it has many, many advantages for a student. [P7]

A teacher added that in nursing and healthcare education, GenAI can support students' practice communication skills by talking with "simulated" patients, helping them gain confidence before encountering real clinical situations: "What I do is that a student has a conversation with a simulant. (. . .) Well, that's a couple of times a year. I would like students to be able to practice at home, and with a simulation program." [P13].

Finally, and in line with the identified opportunities for teaching practice, participants in teacher education highlighted GenAI as a tool that could support students in designing lesson plans and creating learning activities more efficiently. Overall, perceived opportunities for student learning seem to be closely related to both domain-general and discipline-specific forms of knowledge, but the teacher's active role is necessary to guide

students' learning and ensure that they are using GenAI tools thoughtfully and meaningfully to support their learning.

3.2.3. Challenges and Threats

Teachers across disciplines reported multiple perceived challenges and threats associated with the integration of GenAI in their teaching practice. The subtheme mentioned most frequently concerned being overly reliant on GenAI, both by teachers and students, for example, relying on outdated information or content that does not align with the local context. Teachers described several potential threats associated with this reliance, including the erosion of students' learning due to excessive dependence on GenAI or missed opportunities for essential learning experiences. One participant stressed, "the importance of critically evaluating information and avoiding passivity, emphasizing that GenAI should be used as a resource or source of inspiration rather than something to copy without reflection." [P20].

Related to this, another teacher described "info obesity" as a challenge, highlighting the difficulty of teaching students to "discern between valid and invalid information" [P9]. Across the interviews, another frequently raised challenge related to the difficulty of controlling the authenticity of students' work and, in line with this, regulating students' GenAI use by making courses and assessments "AI-proof". One teacher "highlighted the need for greater awareness among teachers," mentioning that without critical reflection, "both students and teachers may rely on GenAI in ways that undermine meaningful learning and assessment practices." [P22].

A (possible) loss of the human aspect of learning and teaching was perceived as another threat by several teachers. They expressed concern that students might start using AI-generated text as a standard, comparing their own work to what the system produces rather than developing their own voice. As one teacher noted: "...if we all start writing in the same way, that personal touch goes completely off." [P23]. This concern extended beyond writing style to the broader teacher–student relationship, with several teachers emphasizing the importance of human interaction in education: "Education is very much about relationships and connections ... even if a student has to work independently, it is a process that should be completed together ... you can't control that by a device or an app, so that is a real threat." [P19].

A final subtheme at the organizational level related to the challenge of staying adaptive as teachers and institutions, including keeping up with GenAI developments and understanding how its integration can be beneficial for specific disciplines. Some teachers also stressed the importance of making sure all teachers are involved via experimentation and training. However, this can be a challenge due to, for example, time constraints or the rapidly changing field. One teacher reflected:

There are several workshops for teachers about how you can use AI. I think it's actually still too little, and I've also noticed that many colleagues are just saying "I don't have the time to figure it all out." (...) I also know that the development is so fast that you start reading and researching now, which will be completely different in three months' time. [P8]

Final concerns and suggestions on how to deal with them related to the transparency of GenAI-generated information and potential bias in the systems. Teachers noted that GenAI tools are trained on large datasets that may contain errors, outdated material, or cultural and regional biases, which can affect both the accuracy of content and students' learning. Also, privacy and data security were mentioned as key concerns by one teacher, particularly regarding how data is stored and who has access.

3.3. Teachers' Perceived Requirements for Responsible, Ethical, and Inclusive Use of GenAI

Teachers identified four essential requirements for professional development and student support: balancing understanding and efficiency when using GenAI tools, critical engagement with GenAI, training on ethical implications, and promoting awareness toward diversity and inclusion. Specifically, most interviewees expressed the need to understand both the functionalities and limitations of GenAI tools. Most teachers mentioned that they must be well-informed about GenAI advancements and usage to guide students responsibly. While teachers recognized the transformative potential of GenAI in education, they also expressed concerns about the lack of transparency, ethical frameworks, and the need for structured institutional support. For example, one teacher mentioned that both colleagues and students are often unaware of the full potential of these technologies and would benefit from structured support [P18], emphasizing the need for clear guidance and accessible resources.

Additionally, a few teachers noted that there is general acceptance that students are already using GenAI tools, and instructors are looking for ways to integrate them meaningfully into their teaching practice. For example, one teacher explained that students are aware of these tools and are already using them but also stressed the importance of contextual guidance. In detail, they mentioned that educators should foster transparency about the use of tools like ChatGPT in their coursework and support students to develop critical thinking around prompting and evaluating GenAI responses [P17]. Because of this, many educators express a need for clearer guidelines and training to help both teachers and students navigate the capabilities, risks, and boundaries of GenAI use. In addition, educators emphasize that while GenAI can assist with administrative tasks, the core responsibilities of teaching and student assessment should remain human-led. One teacher commented that while GenAI tools like autocorrect or language models can assist students, educators must remain focused on what skills and knowledge students need to acquire independently [P12].

Similarly, teachers also stressed the need for accountability in GenAI usage. As concerns about plagiarism and the ethical use of GenAI content arise, educators feel the need to better support students in understanding the importance of intellectual ownership and responsibility. This includes recognizing the potential harm to original authors and the significance of proper attribution. One teacher mentioned that students need to clearly state when GenAI tools have been used for the sake of transparency [P19].

The need for institutional support to better adapt and support students' needs is further reflected in teachers' emphasis on promoting critical thinking among students regarding the use of GenAI. This includes educating students about potential biases and the ethical implications of relying on AI-generated content. As one teacher stressed, "I am going to talk to those students about that. Where does that come from?" [P05], highlighting the emphasis on sense-making and critical evaluation. Professional development should also include programs that address both the pedagogical integration and the ethical considerations surrounding GenAI use.

Along with ethical considerations, teachers emphasize the need for inclusivity in GenAI development and training. They recognized that students from diverse backgrounds may have varying levels of access to technology and GenAI tools. As one teacher pointed out, "if you build something within the paid version (of a GenAI tool), then students who only have the free version can't work with it" [P17]. Moreover, higher education teachers emphasized biases embedded within GenAI systems, which often reflect the values and limitations set by their creators, typically rooted in Western-centric perspectives. They cautioned that these biases could lead to discrimination and unequal access to accurate information, particularly for underrepresented groups like women, migrants, and non-

Western cultures. Teachers ultimately advocated for GenAI education and professional development that promote and support critical thinking, ethical awareness, and equitable access. They also recognized that AI literacy must address who is excluded, who benefits, and who defines the norms embedded in these tools. As one teacher explained, “At the end of the day, you don’t want to create a worse society through an energy system, but rather a better society. So, I think that’s a big challenge” [P07]. All in all, participants highlight that critical and responsible use of GenAI depends not only on technological knowledge but also on pedagogical judgment, ethical awareness, and institutional support.

4. Discussion

As GenAI becomes increasingly embedded in everyday life, both within and outside of educational settings, teachers bring essential expertise to guide decisions on its meaningful integration to enhance teaching and learning. The current study’s findings suggest that teachers’ engagement with GenAI is shaped by an ongoing sensemaking process through which they interpret its implications for teaching and learning, assess potential risks and benefits, and reflect on how it aligns with current teaching practices. Through this process, teachers leverage their pedagogical, content, and technological knowledge to make decisions about when and how to use GenAI in teaching and learning.

The findings reveal variation in teachers’ understanding of GenAI, ranging from basic awareness and describing it in broad terms to more advanced conceptual understanding demonstrated by referring to large language models and differentiating between predictive and generative AI. Teachers’ cautious, exploratory engagement with GenAI aligns with the understanding stage of GenAI integration identified by [Kaplan-Rakowski et al. \(2023\)](#), in which teachers are exploring potential ways to use GenAI before engaging in deeper pedagogical integration. Much of this exploration is motivated by teachers’ efforts to understand students’ use of these tools, resulting in experimenting with free versions with limited functionality. This exploratory engagement furthermore appears to be constrained by a limited understanding of GenAI’s potential for teaching and learning. Consequently, GenAI is perceived as a time-saving tool for efficiency gains rather than a pedagogical resource.

Despite the anticipated efficiencies, teachers expressed frustration that engaging with GenAI requires additional time. This may reflect a transitional phenomenon. Mirroring the information technology productivity paradox of the late 1980s, [Brynjolfsson et al. \(2020\)](#) stressed that complementary innovations, such as new skills, processes, and organizational adaptations, need to be developed and embedded into practice for technological advances to translate into productivity gains. Therefore, the additional time that teachers experience could be a transitional stage in which they are building new skills and exploring how GenAI can be integrated into their workflow. However, the additional time required may also carry cognitive implications. [Rind \(2026\)](#) argues that while AI systems may reduce routine workload and improve efficiency, they come at the cost of displacing teachers from moments of instructional uncertainty during which pedagogical content knowledge is developed. Additionally, [Rind \(2026\)](#) suggests that teachers may experience an ethical–cognitive burden, the need to reconcile their GenAI use with their professional responsibility, accountability, and moral judgment. These tensions are reflected in our findings of how teachers reported their tendency to use GenAI mainly for preparatory activities, such as lesson preparation and development, and less frequently for in-class activities, suggesting limited recognition of GenAI’s potential for human–AI collaboration, individual differentiation, and feedback processes ([Bower et al., 2024](#)). Teachers appear to remain in an experimental phase, characterized by exploration and curiosity, yet without transitioning toward purposeful integration into teaching practice. To support teachers

in adopting GenAI in higher education, there is a need to understand what hinders this transition from exploration to integration.

The shift from exploration to integration of GenAI may be challenging because it is closely linked to broader contextual or systemic factors, which often go beyond individual control. For example, teachers emphasized the need to continuously make sense of GenAI tools while working under pressure created by the quick pace of technological change, lack of time, and limited support structures. The lack of systemic guidance, combined with ethical concerns such as potential biases in GenAI systems, data privacy, and the predominance of Western-centric training data, further increases the demands placed on teachers. Using the TPACK lens, these challenges can be better interpreted. Meaningful integration requires not only technological knowledge about how GenAI tools function, but also teachers' ability to identify how to best integrate GenAI tools with their pedagogical intentions and disciplinary expectations. Teachers' concerns about overreliance, reduced student learning, evaluation of quality, assessment of authenticity and integrity, and loss of personal voice reflect everyday practice-related challenges, highlighting the difficulty teachers face in making pedagogical, technological, and content-related decisions. Concerns about the quality and reliability of GenAI-generated content seem to further complicate this integration process. Our findings suggest that existing TPACK domains alone may not sufficiently capture the demands of GenAI integration. Instead, teachers are required to continually reshape their needs and use of GenAI through ongoing sensemaking. While prior work, including the Intelligent-TPACK (Celik, 2023), has already emphasized the role of ethical and AI-related knowledge, our study further demonstrated that GenAI integration is a dynamic process that involves ongoing, context-dependent negotiation of these knowledge domains in everyday practice.

Contextual knowledge, including ethical guidelines, institutional policies, and resource availability, may add further complexity to teachers' decision-making around GenAI integration. Teachers in this study mentioned the lack of institutional guidance, together with limited professional development opportunities, pointing to misalignment between the knowledge required to meaningfully integrate GenAI and the contextual conditions supporting such integration. This misalignment between innovation and institutional conditions has been well documented in the literature of education innovation implementation (e.g., Georgiou et al., 2025). This misalignment contributes to continued experimentation rather than informed pedagogical change.

Even within this challenging context, our findings suggest that teachers frame their engagement with GenAI as an ongoing sense-making process rather than a linear process of adoption. Although they do not appear resistant to the innovation, they express concerns about appropriate pedagogical, ethical, and assessment-related uses of GenAI (Trikoili et al., 2025), a finding that may explain why many teachers still experimentally engage with GenAI, without moving to an intentional integration in their teaching practice. To support teachers in moving beyond the experimentation phase, GenAI integration needs to be recognized as an ongoing evidence-informed process. Scholarship of Teaching and Learning (SoTL) approaches, where teachers continuously evaluate teaching innovations through their practice, may offer a useful approach to support GenAI integration (Hutchings et al., 2011). SoTL encourages an iterative reflective process of inquiry into how GenAI shapes student learning and assessment, supporting teachers' sense-making and evidence-informed decision-making (Georgiou et al., 2023). Our findings emphasize the importance of providing practice-oriented professional development opportunities that go beyond exploration of GenAI to actual integration. Such opportunities should focus on pedagogical and ethical decision-making and not only on technical skills. In a nutshell, the tension between the potential transformative role of GenAI and the institutional conditions under

which teachers are expected to implement it is multifaceted. It requires not only supporting teachers' technological knowledge but also providing contextual support systems that enable sensemaking, ethical reflection, and evidence-informed teaching practice.

Finally, an interesting recurring theme throughout the interviews consists of whether GenAI is a threat to higher education or whether it can advance education through the opportunities it offers. For teaching, it is valued for inspiration, creativity, reduced workload, and overall efficiency. For students, teachers emphasize opportunities for study support, skill development, and discipline-specific learning. As found in another study by [den Hollander et al. \(2025\)](#), teacher coordinators in Dutch higher education perceived GenAI as an inevitable force. Thus, GenAI can also be seen as an external trigger that requires higher education institutions' adaptive capacity as an entire system ([Louws et al., 2025](#)). Viewing the integration of GenAI into teaching and learning from a systems lens, we could reflect on how to make it work for us as a supportive tool rather than a replacement for human expertise. This is especially relevant in addressing profound, complex educational challenges such as teacher shortages and educational inequalities. As reflected upon by one of the teachers in our study, GenAI can reduce workload, providing more time for the essential student–teacher relationship.

As pointed out by [Louws et al. \(2025\)](#), such educational changes require *collective* sensemaking or a shared understanding of these challenges to avoid friction between actor groups and, consequently, support the system's adaptive capacity building. Building further on our findings and on signals of friction between teachers on whether and how to cope with GenAI in their teaching practice, we advocate, in line with [den Hollander et al. \(2025\)](#), for facilitating structured collective sensemaking sessions via workshops and discussions. These sessions could focus on how GenAI changes both teacher and student roles, and on developing collective solutions for what is required from all educational actors to use GenAI effectively and responsibly. Moreover, we also emphasize the need for diverse training opportunities, ranging from digital literacy to self-regulated learning competencies, for both students and teachers to enable responsible and effective use of GenAI in higher education ([Wong & Viberg, 2024](#)). Especially for beginning teachers, such workshops may be particularly helpful, as adaptive expertise, the ability to flexibly apply knowledge and skills in novel or complex situations, develops through structured professional learning opportunities, reflection, feedback, and experience, which are key elements for becoming adaptive teachers in complex, rapidly evolving technological contexts ([Van Tartwijk et al., 2017](#); [van Bruggen et al., 2026](#)).

Limitations

The current study has several limitations. Firstly, GenAI technologies are evolving at a rapid pace with the development of new features and capabilities. Therefore, teachers' perceptions and uses of GenAI captured in this study may not fully account for new technological developments. As GenAI tools become even more mainstream and more integrated into higher education, teachers' engagement with it will also evolve. Nonetheless, the insights offered by this study into the opportunities and challenges for teaching practice remain relevant as higher education institutions continue to grapple with integrating GenAI responsibly and ethically.

Secondly, the sample size is rather small and consists of teachers from diverse disciplinary backgrounds from one country. The participants were recruited via an open call and email invitation, which may have introduced self-selection bias, as those who chose to participate are likely teachers with greater interest in or awareness of GenAI. Consequently, our findings may not be fully representative of the broader higher education teaching population. Furthermore, a qualitative study means that these findings are based on the

perceptions of the teachers interviewed. However, the focus was not to generalize our results to a broader population, but to offer a detailed and context-specific insight (Polit & Beck, 2010) into the impact of GenAI on teaching practice. Future studies employing random sampling across institutions and disciplines would help to establish whether findings in the current study can be generalised to the wider higher education teaching population.

Finally, the current study employs the TPACK lens to examine how teachers make sense of GenAI for their teaching practice, focusing mainly on the various forms of knowledge required for using GenAI effectively. However, teaching is a complex process and includes motivational factors that have not been fully examined in the current study. Future research could draw on complementary theoretical perspectives, such as the Unified Theory of Acceptance and Use of Technology (Venkatesh et al., 2003) and Self-Determination Theory (Deci & Ryan, 2012), to gain deeper insights into the interplay between contextual and motivational factors of teachers' professional development in the context of GenAI.

5. Conclusions

Teaching and learning practices have continuously evolved and adapted in response to technological advancements throughout history, from the democratization of access enabled by the printing press to the facilitation of learning across space and time by the Internet and digital tools (Yan et al., 2024). The TPACK framework reinforces the indispensable role teachers play in influencing the effective use of technology in education (Mishra et al., 2023). Our work suggests that sensemaking processes play a central role in teachers' engagement with GenAI, since these processes support them in interpreting GenAI's pedagogical value, weighing potential risks, and identifying implications for practice. This phase in history, where educators are cautiously experimenting with GenAI rather than fully integrating it, reflects a period of professional meaning-making instead of pure resistance. Professional learning, collaborative reflection, and clear institutional guidance may be key elements to support teachers' responsible and pedagogically sound integration of GenAI. This is very important, since the potential adoption of GenAI will not only depend on the technology itself but also on how teachers are supported to better understand, evaluate, and adapt this technology in a teaching context.

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Abbreviations

The following abbreviations are used in this manuscript:

TPACK	Technological, Pedagogical, and Content Knowledge
GenAI	Generative Artificial Intelligence
SoTL	Scholarship of Teaching and Learning

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